

Proof-Of-Concept in diagnosing thyroid nodules on ultrasonography using deep CNN in Thailand

Jirapas Wongtreenatrakoon

King Mongkut's University of Technology Thonburi (KMUTT), Thailand

ESIC PLUS Laboratory

Author Note

The suggested idea of including the whole ultrasonography scans as the images used in fine-tuning of the models have resulted in over 4.85% increase in diagnosing performance of the selected models, my appreciation to the professor who brought up the idea which further extended and enriched my study.

As well as to all the seniors and other professors who have helped and guided me along the way.

Proof-Of-Concept in diagnosing thyroid nodules on ultrasonography using deep CNN in Thailand

The goal of this study is to conduct a proof-of-concept, through evaluating the ability of deep convolutional neural network (CNN) in diagnosing thyroid nodules on ultrasonography (US) in term of its performance in differentiating benign (TIRADS 1-3) from malignant (TIRADS 3-5) cases. Such that, if proven, can be applied to benefit the medical fields, specifically radiologists in classifying thyroid nodules, in Thailand. The study is conducted through training and evaluation of 4 selected CNN models namely InceptionResNetV2, AlexNet, InceptionV3, and SqueezeNet. Together forming an ensemble (CNNE), with the predicted value being an average value of all the 4 predicted values. CNNE is trained using data obtained from Kaggle totaling 251 samples each for benign and malignant cases. The study is conducted on both cropped ROI and non-cropped version of the same 502 samples, with the non-cropped version showing over 4.85% increase in diagnosing performance over the cropped version. Throughout both versions, CNNE illustrated high precision (0.88-0.93), recall (0.88-0.93), f1-score (0.88-0.93), and AUC-ROC (0.882-0.925). Overall, CNN shows a satisfactory diagnosis performance in classifying thyroid nodules on US on a binary level and further research and development is in order such that CNN can be applied to aid Thailand's radiologists.

This study is based greatly on a journal article made by (Jieun Koh, 2020) whom and his team conducted a comprehensive training and evaluation on CNNE in binary classifying thyroid nodules on US between benign and malignant. As mentioned, this study acts as a proof-of-concept in the use of CNN in aiding of Thailand's radiologists, an effort to alleviate the current shortage of diagnostic radiologists through reducing their workloads through the application of CNN.

Method

This study utilized training data available on Kaggle (DASMEHDIXTR, 2022), first the images are cropped using the min-max cropping method to cut out the ROI as can be seen in Figure 1.

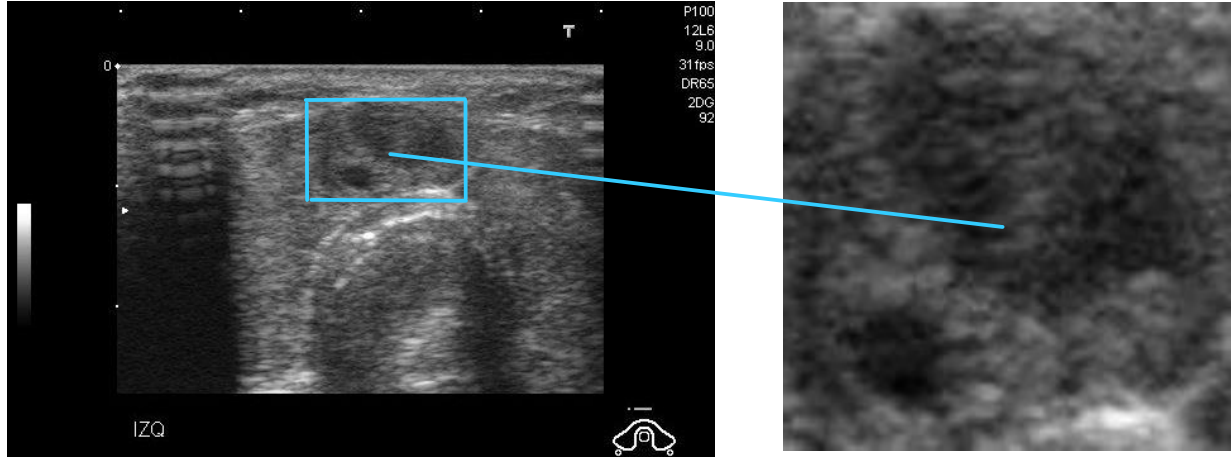


Figure 1: Image processing process, ROI are cropped and adjust the pixel values such that the grayscale of the images lies within the range of 0 to 255, and then images dimensions are adjusted to 224 by 224 through simple linear interpolation between pixels.

After which the normalized images are classified into either benign or malignant. The retrieved training samples come with 251 benign samples and 201 malignant samples. Therefore 50 malignant samples are randomly selected and left-right mirroring images are generated to make up 251 malignant samples in total.

Implementation of CNN models

The 4 CNN models, namely AlexNet, InceptionV3, SqueezeNet, and InceptionResNetV2, are then implemented in Python IDE utilizing the TensorFlow library. Given the small amount of training data, transfer learning is used during the training of the 4 models by freezing the layers of the base model such that the models can utilize their pre-trained shapes. With that the 4 models are fine-tuned using each of the 251 images from the two classes. For the fine-tuning process, 20% of each of benign (0) and malignant (1) images are randomly selected as the validation while the rest of the 80% are used for training. After

which CNNE is ensembled, such that its predicted value (0-1) is obtained from taking the average of the 4 predicted values (0-1) made by the 4 mentioned models.

Statistical Figures of evaluation

Given the constraints in the availability of the images of thyroid nodules on US, the images used for fine-tuning of CNNE is used as evaluating images as well. The statistical figures computed for CNNE are the following: precision, recall, f1-score, and AUC-ROC. The formula for the calculation of each of the figure are as follow:

$$F1 \text{ Score} = 2 * (\text{Precision} * \text{Recall}) / (\text{Precision} + \text{Recall})$$

$$\text{Precision} = \text{True Positives} / (\text{True Positives} + \text{False Positives})$$

$$\text{Recall} = \text{True Positives} / (\text{True Positives} + \text{False Negatives})$$

Information that will be needed to derive those statistical figures:

- True Positives (Instances that are correctly predicted as positive)
- False Negatives (Instances that are positive but are incorrectly predicted as negative)
- True Negatives (Instances that are correctly predicted as negative)
- False Positive (Instances that are negative but are incorrectly predicted as positive)

```
[[219  32]
 [ 27 224]]
Classification Report:
              precision    recall  f1-score   support

     0       0.89         0.87         0.88         251
     1       0.88         0.89         0.88         251

 accuracy          0.88         0.88         502
 macro avg         0.88         0.88         0.88         502
 weighted avg         0.88         0.88         0.88         502

AUC-ROC Score: 0.8824701195219125
```

Figure 2: CNNE's Classification Report, Confusion Matrix, and AUC-ROC Score (Cropped)

As mentioned in the abstract the statistical figures obtained for CNNE are quite satisfactory showing reasonably high precision at 0.88, with the same going for recall and f1-score while having 0.882 as its AUC-ROC score.

However, during the course of this study, it was recommended that the whole original images, as can be seen on the left of Figure 1 be used instead of the cropped versions. Therefore, to test the hypothesis of improved diagnosing performance the same CNNE is fine-tuned again, instead this time using the original images. All the original images have their pixel values adjusted such that the grayscale of the images lies within 0 to 255 and the dimension is shaped into 224 by 224 using simple linear interpolation similar to the study done on the cropped versions. In any case, the total number of samples for each benign and malignant are 241 each as compared to 251 each for the cropped versions. This is due to the manual removal of 10 samples from the benign data pool, given that those 10 samples contain the ROI annotation on them and thus could potentially lead to inaccuracy in comparisons that are going to be made between the two versions. Figure 3 shows the results of the original versions.

```

[[216  25]
 [ 11 230]]
Classification Report:
              precision    recall  f1-score   support

     0           0.95       0.90       0.92         241
     1           0.90       0.95       0.93         241

 accuracy              0.93         482
 macro avg           0.93       0.93       0.93         482
 weighted avg        0.93       0.93       0.93         482

AUC-ROC Score: 0.9253112033195021

```

Figure 3: CNNE's Classification Report, Confusion Matrix, and AUC-ROC Score (Uncropped)

In comparison to the cropped versions there is observed an across-the-board improvement in precision, recall, and f1-score, all at 0.93 in contrast to the 0.88 of the cropped versions. While the AUC-ROC score improved from 0.882 to 0.925 accounting for over 4.85% improvements from the cropped

versions. This finding can potentially be explained by the idea that, by using the original images to fine-tune CNNE this provides the ensemble with the whole US instead of just the thyroid nodules themselves which allows it to find patterns upon the whole US and not just the thyroid nodules. In other words, this means that there could be specific patterns exist outside the nodule itself that constitute to whether a thyroid nodule is considered benign or malignant and therefore should be considered as well during future training and predicting processes.

Discussion

In this study, even though the evaluation samples used are the same as the ones used for fine-tuning which may explain the relatively high AUC-ROC score achieved for both versions of CNNE, being in the high range of 0.882-0.925, nonetheless valuable insights were derived. For instance, insight into the possibility for there to be patterns outside the thyroid nodules which can be used as a deciding factor to whether a thyroid nodule is deemed benign or malignant. Insight of which stands as a proof for CNN's capability in diagnosing thyroid nodules on US, and that further research and development with greater and more reliable data pools on various other CNN models should be conducted such that Thailand's field of radiology can stand to benefit from the integration of this technology.

References

DASMEHDIXTR. (2022). *DDTI: Thyroid Ultrasound Images*. Retrieved from Kaggle:

<https://www.kaggle.com/datasets/dasmehdixtr/ddti-thyroid-ultrasound-images/data>

Jieun Koh, E. L.-K.-M.-r. (2020). Diagnosis of thyroid nodules on ultrasonography by a deep convolutional neural network. *Nature Research*, 1-7.